Elementary & Intermediate Algebra

ANSWER KEY

Solving linear equations

1 Solve for x: $4x - 9 = 23$.

$4x = 32$, so $x = 8$.

2 Solve for x: $\frac{3x - 2}{5} = 8$.

Multiply both sides by 5: $3x - 2 = 40$, so $3x = 42$, giving $x = 14$.

Solving and graphing inequalities

3 Solve $5x + 3 < 28$, giving your answer as an inequality.

$5x < 25$, so $x < 5$.

4 Solve $-4x + 7 \leq -9$, being careful with the direction of the inequality.

$-4x \leq -16$. Dividing by -4 flips the inequality: $x \geq 4$.

Systems of linear equations

5 Solve the system: $y = 3x - 4$ and $2x + y = 11$.

Substitute: $2x + (3x - 4) = 11$, so $5x - 4 = 11$, giving $5x = 15$, $x = 3$. Then $y = 3(3) - 4 = 5$. Solution: $(3, 5)$.

6 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$, using elimination.

Multiply the second equation by 2: $2x - 4y = -8$. Add to the first: $5x = 10$, so $x = 2$. Substituting into $x - 2y = -4$: $2 - 2y = -4$, so $-2y = -6$, giving $y = 3$. Solution: $(2, 3)$.

Factoring and solving quadratic equations

7 Solve $x^2 - x - 12 = 0$ by factoring.

Factor: $(x - 4)(x + 3) = 0$, so $x = 4$ or $x = -3$.

8 Solve $2x^2 - 5x - 3 = 0$ by factoring.

Factor: $(2x + 1)(x - 3) = 0$, so $x = -\frac{1}{2}$ or $x = 3$.

Simplifying rational and radical expressions

9 Simplify $\frac{x^2-9}{x+3}$.
Factor the numerator: $\frac{(x-3)(x+3)}{x+3} = x-3$ (for $x \neq -3$).

10 Simplify $\sqrt{48}$.
$\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3}$.

Working with exponents

11 Simplify $\frac{x^7}{x^3}$.
$\frac{x^7}{x^3} = x^{7-3} = x^4$.

12 Simplify $(3x^2y)^3$.
$(3x^2y)^3 = 3^3x^6y^3 = 27x^6y^3$.

Absolute value equations

13 Solve $|3x+2| = 11$.
Either $3x+2 = 11$, giving $x = 3$, or $3x+2 = -11$, giving $x = -\frac{13}{3}$.

Word problems leading to linear or quadratic equations

14 The length of a rectangle is 3 more than twice its width, and its area is 65. Find the width and length.
Let width $= w$, length $= 2w+3$. Area: $w(2w+3) = 65$, so $2w^2+3w-65 = 0$. Factor: $(2w+13)(w-5) = 0$, giving $w = 5$ (rejecting the negative solution $w = -6.5$). Length: $2(5)+3 = 13$.

15 A number increased by 8 is 3 times the original number decreased by 4. Find the number.
Let the number be x: $x+8 = 3x-4$. Solving: $12 = 2x$, so $x = 6$.

Combining algebraic techniques in a multi-step problem

16 This question has two parts. Two consecutive positive integers have a product of 132. (a) Set up and solve a quadratic equation to find the two integers. (b) Verify your answer by checking the product.
(a) Let the integers be n and $n+1$: $n(n+1) = 132$, so $n^2+n-132 = 0$. Factor: $(n+12)(n-11) = 0$, giving $n = 11$ (rejecting the negative solution $n = -12$). The integers are 11 and 12. (b) Check: $11 \times 12 = 132$ ✓.

Solving quadratics using the quadratic formula

17 Solve $x^2 + 3x - 2 = 0$ using the quadratic formula, giving exact answers.
$x = \frac{-3 \pm \sqrt{9+8}}{2} = \frac{-3 \pm \sqrt{17}}{2}$.

18 Use the discriminant to determine the number of real solutions of $4x^2 - 12x + 9 = 0$, without solving.
Discriminant $= (-12)^2 - 4(4)(9) = 144 - 144 = 0$. Since the discriminant is zero, there is exactly one real (repeated) solution.

Factoring by grouping and special patterns

19 Factor $x^3 + 2x^2 - 9x - 18$ by grouping.
Group terms: $x^2(x+2) - 9(x+2) = (x+2)(x^2 - 9)$. Factor the difference of squares: $(x+2)(x-3)(x+3)$.

20 Factor $4x^2 - 25$ completely.
Difference of squares: $4x^2 - 25 = (2x-5)(2x+5)$.

Rational equations and extraneous solutions

21 Solve $\frac{x}{x-2} = \frac{3}{x-2} + 1$, checking for extraneous solutions.
Multiply through by $(x-2)$: $x = 3 + (x-2)$, so $x = x + 1$, giving $0 = 1$ — a contradiction, meaning there is no solution (this reflects that $x = 2$ would make the denominators undefined, and no other value works).

Multi-step algebraic word problems

22 This question has two parts. A boat travels 60 miles downstream in 3 hours and returns upstream in 5 hours. (a) Let b be the boat's speed in still water and c be the current's speed; write two equations based on the trip times. (b) Solve the system to find b and c.
(a) Downstream speed is $b + c$: $3(b+c) = 60$, so $b + c = 20$. Upstream speed is $b - c$: $5(b-c) = 60$, so $b - c = 12$. (b) Adding the two equations: $2b = 32$, so $b = 16$. Then $c = 20 - 16 = 4$. The boat's speed in still water is 16 mph, and the current's speed is 4 mph.